

two facilities or providers—one sending site and one receiving site—and provides HIPAA-compliant disaster recovery with three redundant copies of each patient record. Facilities can add additional clinical specialties as needed. A mini-clinic configuration is available for small sites or group practices. Medweb's Web-based solution offers computer-based patient assessment protocols and integrates medical imaging, information management and security.

Such fully-managed telemedicine services are still a rarity in India, and provide a huge opportunity for start-ups, as it would make it extremely easy for hospitals and other healthcare institutions to expand their reach. The market is likely to be extremely large, once the concept gains traction.

Portable and network-capable medical equipment

Telemedicine is not just about information technology and connectivity. It is also about the right medical equipment—portable, network-capable and robust instruments. A lot of work is happening towards that as well. The availability of extremely miniaturised and highly-capable lab-on-a-chip modules has indeed taken telemedicine forth significantly. The development of full-featured yet tiny medical equipment is a constant challenge posed by chipmakers like Intel to engineers during various embedded system design contests.

One recent example of such a device is the invention of the world's smallest and lightest telemedicine microscope, announced in April. Aydogan Ozcan of the University of California, Los Angeles, unveiled details of his microscope in a paper published online in the journal *Lab on a Chip*. The instrument builds on an imaging technology developed by him, known as LUCAS (Lens-less Ultra-wide-field Cell Monitoring Array platform based on Shadow imaging). Instead of using a lens to magnify objects, LUCAS

generates holographic images of micro-particles or cells by employing a light-emitting diode to illuminate the objects and a digital sensor array to capture their images.

Weighing 46 gm, the microscope is a self-contained imaging device. The only external attachment required is a USB connection to a smartphone, PDA or computer, which supplies the microscope with power and allows images to be uploaded for conversion into results that are then sent to a hospital. Samples are loaded using a small chip that can be filled with saliva or a blood smear for health monitoring. With blood smears, the lens-less microscope is capable of accurately identifying cells and particles, including red blood cells, white blood cells and platelets. The technology has the potential to help monitor diseases like malaria, HIV and tuberculosis.

The inventor believes that tools like the lens-less microscope could be digitally integrated as part of a telemedicine network that connects various mobile healthcare providers to a central lab or hospital, filling gaps in physical infrastructure with mobile tools. Transmission connections for such networks already exist in cellular networks, which have penetrated even the remotest corners of the globe. The microscope is also very cost-effective, allowing mass deployment even in third-world nations.

Robots in telemedicine

The Mayo Clinic, a well-known non-profit healthcare institution in the US, has debuted 'robot doctors' this year. This ensures that patients in less densely-populated areas also have immediate access to stroke expertise and clot-busting medication necessary for treatment. When an emergency arises, the rural hospital places a call to an associated medical centre (or hub). The hub vascular neurologist is equipped with a telemedicine tool belt that includes a smart phone with a tele-radiology application and other technology such as a laptop with a Web camera.

The remote, spoke hospital is equipped with a mobile robotic telemedicine camera system that is positioned near the patient's bed. From afar, the stroke physician can observe and speak with the patient, and healthcare providers do a real-time consultation and review CT scans of the brain. If a stroke diagnosis is confirmed, appropriate treatment can be quickly administered, such as a clot-busting drug when a clot is blocking blood flow to the brain.

Similarly, the Frankfort Regional Medical Centre in Kentucky now relies on experts from the University of Louisville Hospital's stroke centre. A robotic telemedicine machine placed in an examination room sees, hears and diagnoses patients, then suggests treatment.

Note that a robot in telemedicine does not have to be something hi-fi used to remotely control and perform surgeries. It could mean a simple computerised figure that sees, hears and speaks to patients, controlled by doctors through a network.

Even from this small sample of developments, it is obvious that telemedicine is an evolving technological space. Several improvements in the form of advanced, network-capable medical equipment, wireless communication technologies, mobile services, data acquisition, interpretation and archival systems, etc have given the field a boost. However, challenges and hence opportunities abound in India, to develop more user-friendly technologies, link up the last mile with reliable communication facilities, foster confidence amongst the people to go for telemedicine services and so on. With such a large population at the bottom of the pyramid without access to hospitals and, ironically, a well-matched population in the cities that is reluctant to go to the hospital even though there exists one, there is a business case strong enough for tech-makers to participate in the telemedicine revolution! ●

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